MASKINHALLEN · USER MANUAL · VERSION 2.0

FASFEL

PHASE-ERROR MACHINE · KNOCKS · CLICKS · METAL

Dry wooden, metallic and glassy knocks that sit above your drums and bass, played against themselves until they come alive.



VST3 · AU · LV2 · STANDALONE
MACOS · WINDOWS · LINUX

MASKINHALLEN

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01 WELCOME

FASFEL is a phase-error machine for hard and minimal techno. No melodies, no bass, no pads: just the dry wooden, metallic and glassy knock that sits above the groove.

A short physical impulse strikes a body, a resonator or one of eight engines, and FASFEL plays a copy of it against itself: flipped, detuned and shifted, so the two knock against and cancel each other. That phase error is what makes the knock come alive. It plays by itself in time with your DAW, or from MIDI.

- **Eight engines in two layers**, blended with MORF.
- **FASFEL, FASDRIFT, FASVÄND**: the phase machinery.
- **A skewed euclidean sequence** with two rings that drift apart.
- **The map** between TORR, SNED, FULL and KALL.
- **MÖRKERSTEGET** (the darkness stage): rumble, pressure, darkness.
- **362 presets** and your own sounds.

SWEDISH LABELS

`Spela / Djup`	Play / Deep (tabs)	`Kropp / Slag`	Body / Strike
`Färg / Form / Fall`	Colour / Shape / Decay	`Skev / Takt`	Skew / Rate
`Muller / Tryck / Mörker`	Rumble / Pressure / Darkness	`Botten`	Bottom (high-pass)
`Spela själv`	Play by itself	`Frö`	Seed
`Kast / Knuff / Ångra`	Throw / Nudge / Undo	`Ladda eget`	Load your own

THE WINDOW



02 INSTALL

Unzip the file for your system. You get a folder called **FASFEL** (on Mac and Windows: FASFEL-macOS and FASFEL-Windows). Then run the installer for your system and restart your DAW (or rescan plug-ins). FASFEL shows up as an instrument.

MACOS

1. Double-click `FASFEL-macOS.zip` in Downloads.
2. Open Terminal and run:
   `bash ~/Downloads/FASFEL-macOS/installera.command`
3. Quit your DAW completely and start it again.

VST3, AU and a standalone app. macOS 11 or newer, Intel or Apple Silicon. The installer copies the plug-ins and removes macOS's block on downloaded plug-ins.

WINDOWS

1. Right-click `FASFEL-Windows.zip` and choose **Extract all**.
2. Open the `FASFEL-Windows` folder and right-click `installera.bat` → **Run as administrator**.
3. If Windows says "Windows protected your PC": **More info** → **Run anyway**.

VST3 (in C:\Program Files\Common Files\VST3) and a standalone .exe. 64-bit Windows 10 or 11.

LINUX

1. Unzip `FASFEL-Linux.zip`.
2. Run `bash ~/Downloads/FASFEL/installera-linux.sh`. It copies the plug-ins to `~/.vst3` and `~/.lv2`.
3. Rescan plug-ins in your DAW (Bitwig, Reaper, Ardour and others).

VST3 and LV2, 64-bit x86 Linux (built on Ubuntu 22.04).

> SWEDISH OR ENGLISH
>
> The machines have Swedish names on the panel. One click on `SV | EN` in the header switches labels, menus and tooltips to English. Your choice is remembered for every MASKINHALLEN machine.

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03 UNLOCK WITH YOUR KEY

Your purchase comes with a licence key: four groups of four characters, like `XXXX-XXXX-XXXX-XXXX`. You find it in the Payhip email and on your receipt.

1. Open FASFEL. Until it is unlocked, the window shows **UNLOCK FASFEL**.
2. Paste the whole key in the field and press `UNLOCK`.
3. Done. You only need to do this once per computer: every MASKINHALLEN machine on the computer finds the key. A bundle key unlocks every machine.

The key is not case sensitive, and I/L and O/0 are treated as the same character, so a typo there does not matter. If it says the key does not work, check that the whole key was pasted. You can use your key on your own computers.

THE FREE DEMO

Every machine has a free demo with the same sound engine. Some presets and functions are locked, and the demo has its own name (FASFEL DEMO), so it can sit next to the full version. Buying the full version is what unlocks everything; install it and paste your key.

UPDATES

Updates are free for life. Download the new zip from your Payhip receipt link and run the installer again. Your projects and your key are kept.

04 YOUR FIRST MINUTE

1. Put FASFEL on an instrument track and press play. `SPELA SJÄLV` is on from the start, so it plays in time right away.
2. Step through presets. Start in MINIMAL, METALL or FASFEL.
3. Turn `FASFEL` and `FASDRIFT` and listen to the knock start to move against itself.
4. Press `SLUMPA LJUD` for a new sound from the recipes, or `KNUFF` to move everything a little. `ÅNGRA` goes back.
5. Put it over BETONG and GRUND. The low end below 150 Hz is always mono, and the bottom high-pass keeps it out of the kick's way.

05 THE EIGHT ENGINES

Two layers, **A** and **B**, each play one engine. `MORF` blends them; `MORFSLUMP` lets every hit land on a new spot between them. `FÄRG` (colour), `FORM` (shape) and `FALL` (decay) control both.

`KROPP`	Body: modal resonators (marimba, wood block, bells, glass, gong, anvil, stone and more, 24 bodies), or the map's overtones.
`KROCK`	Crash: sine, triangle or folded wave with a pitch drop: kick, tom, zap.
`VRID`	Twist: phase distortion, from saw to a resonant sweep.
`FM`	Two operators with feedback.
`KAM`	Comb filter with plus or minus feedback.
`SLINGA`	Loop: an all-pass loop, laser and spring.
`KORN`	Grains from RÖR, MOTOR, BRUS, RÖST, KEDJA, GLAS, BETONG, ELNÄT, or your own sound (EGET).
`LÅS`	Lock: a phase-locked loop that loses its grip.

YOUR OWN SOUND

Press `LADDA EGET` or drag a wav, aiff or flac onto the window. KORN cuts grains from it.

06 THE PHASE MACHINERY

`FASFEL`	Phase error: each engine plays a copy of itself, flipped, detuned and shifted, so they knock against and cancel each other.
`FASDRIFT`	Phase drift: the shift wanders in time, from one turn per 16 bars to two per beat, plus a moving comb on the right channel.
`FASVÄND`	Phase flip: stereo through phase. Below 150 Hz always mono.

The **phase scope** in the window shows the stereo image, with a correlation meter.

07 THE SKEWED SEQUENCE

FASFEL plays by itself in your DAW's tempo (`SPELA SJÄLV`). MIDI in works at the same time.

- **Ring A** (orange) and **ring B** (blue): euclidean rhythms, for example 5 of 16 against 2 of 7. They slide against each other, so the pattern almost never repeats. Ring B plays a fifth up.
- `TAKT`: 1/16, 1/16T, 1/8, 3/16, 5/16 or 1/32.
- `SKEV`: every hit comes a little late (never early), and odd steps drag a little extra.
- `TON` and `SPRIDNING`: root note and how much the notes jump (fifth, octave, minor).
- The same `FRÖ` (seed) gives the same pattern. The dice randomises the rings.

08 THE DJUP TAB: MAP, STRIKE AND TUNING

THE MAP

Drag the dot between four corners. The knock changes character and overtone series.

CORNER	CHARACTER	OVERTONES
`TORR`	Dry: dub echo, a single ping	Natural series (1, 3/2, 2, 3)
`SNED`	Skewed: chaotic timing, dust, drift	11th, 7th, 13th harmonics
`FULL`	Drunk: a glass drop, drunken swing	Free wooden bar 1 : 2.756 : 5.404
`KALL`	Cold: comb filter, vacuum suction, precision	Drum skin 1 : 1.593 : 2.136

STRIKE AND TUNING

- `SLAG`: how the body is set going: hammer, rubber, felt, nail, brush, stone, stick, finger, bounce or scrape.

- Scales: 12-TET, 24-EDO, just intonation and slendro. The golden ratio rotates the timbre every hit, so repetition stays alive.
- `TRYCK` (a falling tone under the knock), `SVANS` (tail, up to five times longer), `RÖSTER` (1–6 voices) and `VANDRA` (chance that a hit takes another body).

MIDI AND THE KICK

- The kick note (C1) **or** a sidechain signal (for example BETONG) chokes the tail.
- `GHOST GATE` 0–120 ms: hits that land on the kick wait until the kick has passed.
- `KANAL`: listen to one MIDI channel, for example the one GLAPP sends on.

09 MÖRKERSTEGET AND THE SAFETY RAIL

`MULLER`	Rumble: a dark concrete room under the knock, ducked.
`TRYCK`	Pressure: parallel compression and saturation.
`MÖRKER`	Darkness: takes the top off.
`BOTTEN`	The high-pass, 60–600 Hz. It can also be switched off completely.

A soft limiter keeps the output safe.

10 PRESETS

362 presets: the original 40 (ten per map corner) plus 322 in TUNG, MINIMAL, FASFEL, METALL, MASKIN, SLINGOR and BOTTEN. `SLUMPA LJUD` builds a new sound from the recipes, `KNUFF` moves everything a little and `ÅNGRA` goes back. The dice in each section randomises just that section.

11 TROUBLESHOOTING

FASFEL does not show up	Quit the DAW completely and start it again, or rescan plug-ins. On Mac, run the installer again: it removes macOS's quarantine on the plug-in. On Windows, check that `FASFEL.vst3` is in C:\Program Files\Common Files\VST3.
macOS says the plug-in is damaged	That is macOS's block on downloaded files. Run `bash ~/Downloads/FASFEL-macOS/installera.command` again.

Windows protected your PC	Click **More info** → **Run anyway**. The plug-ins are not signed with a paid certificate.
The key does not work	Paste the whole key, including the dashes. A key for one machine only unlocks that machine; a bundle key unlocks all.
No sound	Press play in the DAW: the machines follow the transport. Instruments go on an instrument track, effects on an audio track or bus.

Anything else? Ask through the store, and the person who built the machine answers.

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